Prompt Radiation Utility vs. Excel Template Comparison

Comparison of new Qt utility output with PromptRadiationTemplate_Expanded_FY2027 Excel calculation

1. Executive summary

Main result: The new utility gives N0318 = 21.49 mrem/h, while the Excel template gives approximately 19.3 mrem/h. The utility is higher by 2.19 mrem/h, or +11.4%. This is a reasonable first comparison for the intermediate text-file version, but it is not yet a bit-for-bit reproduction of Excel.

Most likely cause of the N0318 difference: the low-ion block used by the utility is not the same as the low-ion block shown in Excel. In the utility output, He-4 low and H-3 low alone contribute about 5.26 mrem/h to N0318. In the Excel pasted low-ion block, the low-ion subtotal is about 1.6 mrem/h. This difference can explain most of the N0318 offset.

Secondary issue: some monitors with large P5 sensitivity, especially N0314, N0305, and N0308B, are approximately 29-30% lower in the utility. This suggests that the detailed isotope rows, low-ion locations, or DB0.x location assignment should be checked carefully against the Excel sheet.

Item	Utility	Excel / observation	Interpretation
N0318 total	21.494 mrem/h	19.3 mrem/h	+2.194 mrem/h (+11.4%)
Largest absolute differences	N0314, N0305, N0308B	Excel is higher	Utility is about 70% of Excel for these monitors
Largest likely source of N0318 mismatch	Low-ion block	Utility low-ion assumptions differ from Excel	Align low-ion rows first
Input-file run status	70 yield rows accepted/skipped 27; 97 position rows accepted/skipped 0; 60 detailed + 5 low-ion after cutoff	From utility output	Check whether skipped yield rows are expected

2. Monitor total comparison

Excel totals were copied from the spreadsheet in scientific notation and appear rounded to approximately three significant digits. Therefore, very small differences should not be over-interpreted; the important differences are the systematic 10-30% shifts.

Monitor	Utility mrem/h	Excel mrem/h	Utility - Excel	Difference %	Utility / Excel	Comment
N0318	21.4941	19.3	+2.19415	+11.4%	1.114	Utility higher
N0306	0.823863	1.05	-0.226137	-21.5%	0.785	Utility lower
N0319	1.11408	1.57	-0.455918	-29.0%	0.710	Utility lower
N0325A	121.563	114	+7.56256	+6.6%	1.066	Good agreement
N0325B	16.2655	15.1	+1.16549	+7.7%	1.077	Good agreement
N0304B	44.3522	52.7	-8.34781	-15.8%	0.842	Utility lower
N0314	1331.43	1900	-568.569	-29.9%	0.701	Utility lower
N0305	1331.43	1900	-568.569	-29.9%	0.701	Utility lower
N0308B	744.844	1050	-305.156	-29.1%	0.709	Utility lower
N0322	0.813008	0.921	-0.107992	-11.7%	0.883	Utility lower
N0323	1.02918	1.12	-0.0908195	-8.1%	0.919	Good agreement
N0324	0.0977429	0.0806	+0.0171429	+21.3%	1.213	Utility higher
N0304	0.0694394	0.0984	-0.0289606	-29.4%	0.706	Utility lower
N0315	0.181308	0.15	+0.0313082	+20.9%	1.209	Utility higher

3. Top N0318 contributors from the utility output

The first two utility contributors are low-ion rows. Their combined contribution is about 5.26 mrem/h, which is already larger than the complete Excel low-ion subtotal shown in the pasted table.

Rank	Fragment	Location	DB0.x [mm]	Rate [pps]	N0318 [mrem/h]
1	He-4 low	P2	--	2.35e+09	2.91665
2	H-3 low	P2	--	1.01e+09	2.3435
3	36P	P2	--	5.56e+08	1.65622
4	28Mg	P4	-140.73	2.82e+08	1.27688
5	26Na	P4	-168.16	2.64e+08	1.20704
6	23Ne	P3	-232.27	2.39e+08	1.08217
7	21F	P3	-262.66	2.05e+08	0.932121
8	34Si	P2	--	2.58e+08	0.779347
9	25Na	P3	-203.66	1.65e+08	0.743427
10	24Ne	P3	-197.69	1.42e+08	0.705502

4. Low-ion block comparison

Important observation: The utility and Excel are not using the same low-ion rows. This is the clearest explanation for the N0318 difference. Before checking deeper formulas, make the low-ion rows identical in both calculations.

Excel low-ion block visible in the pasted table:

Ion	A	Z	pps	MeV/u	Scale	Loc	N0318 factor	N0318 mrem/h
H-3 low	3	1	6.5e+08	120	0.524	P2	0.16	0.544
H-2 low	2	1	8.5e+08	80	0.117	P3	0.29	0.288
H-1 low	1	1	3e+09	100	0.0427	P2	0.16	0.205
He-4 low	4	2	2e+08	200	0.682	P3	0.29	0.396

Utility low-ion assumptions currently used or implied by the output:

Ion	A	Z	pps	MeV/u	Loc	N0318 mrem/h
H-3 low	3	1	1.01e+09	218.51	P2	2.3435
H-2 low	2	1	1.63e+09	223.4	P5	~0.16
He-4 low	4	2	2.35e+09	215.65	P2	2.91665
H-1 low	1	1	0	0	P4	0
He-3 low	3	2	7.5e+07	214.63	P3	~0.09

Low-ion conclusion: Excel low-ion subtotal is about 1.6 mrem/h for N0318. The utility low-ion block is approximately 5.5 mrem/h for N0318 using the current default rows. The difference is about +3.9 mrem/h, larger than the total N0318 utility-vs-Excel difference of +2.19 mrem/h. That means the low-ion mismatch is dominant, but the detailed non-low rows are probably lower in the utility by roughly 1.7 mrem/h and should also be checked.

5. Conclusions and recommended checks

Priority	Recommendation
Do first	Make the utility low-ion rows identical to the Excel low-ion block. Use the same rates, energies, locations, and light-ion multiplier. Then rerun the comparison.
Check DB0.x matching	Several top contributors have DB0.x shown as "--" and therefore fall back to P2. Confirm this matches the Excel VLOOKUP behavior and not a fragment-name mismatch.
Check skipped rows	The utility reports 70/27 yield rows accepted/skipped, but only 60 detailed rows after cutoff. Confirm whether the 27 skipped rows are blank/invalid and whether the 10 removed by cutoff are expected.
Check P5-sensitive monitors	N0314, N0305, and N0308B are about 29-30% lower than Excel. Because these monitors have very large P5 factors, small differences in P5 assignment or low-ion location can produce large differences.
After alignment	A good target for this intermediate utility is agreement within a few percent for N0318 and comparable agreement for all monitor totals, limited by Excel rounding and any intentional changes in assumptions.

Prepared from the utility output and Excel values pasted in the comparison request. Excel totals are rounded as pasted; use exact cell values from the workbook for final numerical validation.